

MGT 404 Review Session Problems

Tax

A group of local activists in Cambridge, Massachusetts is trying to encourage lawmakers to expand the city's use of taxes to reduce consumption of products with negative externalities. Specifically, the group believes that the city should tax beer at restaurants, reasoning that higher beer prices would mean fewer drunk college students, who (they argue) make a lot of noise and upset downtown residents. The group favors imposing a tax of \$2 per beer served at restaurants. For this question assume there is no baseline alcohol tax.

Cambridge restaurants provide beer according to the following weekly supply curve:

$$P_S(Q) = 2 + \frac{1}{5}Q,$$

and face the following weekly demand curve:

$$P_D(Q) = 12 - \frac{1}{5}Q,$$

where Q is in 1,000s of bottles per week.

a) What is the market equilibrium price of beer sold in a restaurant in Cambridge and the total quantity sold per week with no tax?

Solution:

We can solve this by first equating the prices and then solving for the equilibrium quantity:

$$\begin{aligned}P_S(Q) &= P_D(Q) \\2 + \frac{1}{5}Q &= 12 - \frac{1}{5}Q \\ \frac{2}{5}Q &= 10 \\ Q^* &= 25\end{aligned}$$

Given the units, this means that 25,000 bottles of beer are sold each week in Cambridge. Evaluating the market demand at $Q^* = 25$ gives us the equilibrium price:

$$\begin{aligned}P_D(25) &= 12 - \frac{1}{5}(25) \\ &= 12 - 5 = 7\end{aligned}$$

So the equilibrium price is \$7 per bottle.

b) What would the new market equilibrium be if the beer tax were in place?

Price (inclusive of tax): _____

Quantity: _____

Solution:

We first shift the supply curve by the tax:

$$P_S^{tax}(Q) = 2 + \frac{1}{5}Q + 2 = 4 + \frac{1}{5}Q$$

and then equate the prices and solve for the equilibrium quantity:

$$P_S^{tax}(Q) = P_D(Q)$$

$$4 + \frac{1}{5}Q = 12 - \frac{1}{5}Q$$

$$\frac{2}{5}Q = 8$$

$$Q^{*,tax} = 20$$

So there are now 20,000 bottles of beer sold per week with the tax in place. The new market equilibrium price is now \$8 per bottle:

$$4 + \frac{1}{5}(20) = 8$$

c) What is the total tax revenue? What percent of the tax is borne by consumers?

Total tax revenue) (\$): _____

% paid by consumers (\$): _____

Solution:

The total tax revenue (in 1,000s) is:

$$tax\ per\ bottle * Q^{*,tax} = 2 * 20 = 40$$

So the government brings in \$40,000 per week from this tax.

The percent paid by consumers is:

$$\frac{P_{cons}^{*,tax} - P^*}{P_{cons}^{*,tax} - P_{supplier}^{*,tax}}$$
$$\frac{8 - 7}{8 - 6} = 0.5$$

So the consumer pays 50% of the tax.

Production

Baking up some profits. Crumbs Bakery, which has been in the news lately, sells cupcakes through retail stores. The production of cupcakes takes both raw ingredients, and labor. The daily production function for cupcakes is given by

$$q = \frac{1}{2} \cdot r \cdot l + l^2 - 20$$

Where output is in dozens of cupcakes, r is pounds of raw ingredients, and l is hours of labor. The cost of a pound of raw ingredients is $p_r = \$1$ while the cost of an hour of labor is $p_l = \$12$.

(a) (8 points). What is the optimal number of pounds of raw ingredients to use per hour of labor?

Pounds of raw ingredients per hour of labor: 8 _____

Answer

The marginal products of raw ingredient and labor are given by

$$MP_r = \frac{\partial q}{\partial r} = \frac{1}{2}l$$

$$MP_l = \frac{\partial q}{\partial l} = \frac{1}{2}r + 2l$$

with marginal productivity of investment

$$MPL_r = \frac{MP_r}{p_r} = \frac{\frac{l}{2}}{1} = \frac{l}{2}$$

$$MPL_l = \frac{MP_l}{p_l} = \frac{\frac{1}{2}r + 2l}{12} = \frac{r + 4l}{24}$$

To solve for the optimal mix of inputs, we equate the marginal productivities of investment for all our inputs:

$$MPL_r = MPL_l$$

$$\frac{l}{2} = \frac{r + 4l}{24}$$

$$12l = r + 4l$$

$$r = 8l$$

So Crumbs should use 8 pounds of raw ingredients for every hour of labor.

(b) (7 points). A standard day requires about 25 dozen cupcakes to be made. What is the cost-minimizing production plan for 25 dozen cupcakes? How much will this cost?

Pounds of raw ingredients: 24_____

Hours of labor: 3_____

Total cost: \$60_____

Answer

If Crumbs wants to make 25 dozen cupcakes,

$$25 = \frac{1}{2} \cdot r \cdot l + l^2 - 20$$

rearranging and plugging in the expression for the optimal mix of raw ingredients and labor gives

$$45 = \frac{1}{2} \cdot (8l) \cdot l + l^2 = 4l^2 + l^2 = 5l^2$$

$$l^2 = 9$$

$$l^* = 3$$

and so we have $r^* = 8 \times l^* = 24$. The total cost function is

$$TC = p_r r + p_l l$$

plugging in the prices and optimal l^* and r^* gives the cost of producing 25 dozen cupcakes:

$$TC = 24 + 12 \times 3 = 60$$

Perfect Competition

1. *Medical Devices.* Jaysun Glove is a small manufacturer and distributor of medical examination gloves. The production process of such latex gloves is highly standardized and the industry is very competitive. Operating an assembly line to produce the gloves has a fixed cost of \$20,000 per month, and the marginal cost of a case of 1,000 gloves increases with the scale of production. The total cost (in dollars) of producing q cases of gloves in a month and per assembly line is:

$$TC(q) = 20000 + 0.005 \cdot q^2$$

- a. Find the marginal cost function, $MC(q)$.

$MC(q)$: _____

$$MC(q) = 0.005 \cdot 2 \cdot q = 0.01q$$

- b. Find the average cost function, $AC(q)$.

$AC(q)$: _____

$$AC(q) = \frac{TC(q)}{q} = \frac{20000}{q} + 0.005q$$

- c. Assume that the market has come to a long-run (allowing for entry and exit) equilibrium. What is the price of a case of gloves?

Price: _____

Answer: Min of ATC is found at the Q where MC=ATC

$$0.01q = \frac{20000}{q} + 0.005q$$

$$0.005q^2 = 20000$$

$$q^2 = 4000$$

$$q = 2,000$$

$$p = 20$$

Price Discrimination (1)

Pricing at the Taj Mahal. The Taj Mahal in Agra is the most popular monument in India, attracting thousands of visitors daily. The total daily demand for visits from foreign tourists is:

$$Q^F(p) = 4,000 - 200p,$$

where p is measured in US dollars. The Taj Mahal's operator, the Agra Development Authority (ADA), incurs a cost of \$0.50 for every visitor who tours the Taj Mahal.

- (a) (4 points). Find the profit-maximizing price that the ADA should charge to foreign tourists. How much profit does the ADA make every day?

Optimal price: \$10.25

Daily profit: \$19,012.50

Answer:

$$Q^F(p) = 4,000 - 200p \rightarrow P = 20 - 0.005Q$$

$$\pi = (20 - 0.005Q)Q - 0.5Q$$

$$\frac{d\pi}{dQ} = 20 - 0.01Q - 0.5 = 0 \rightarrow Q = 1950$$

$$P(1950) = 10.25$$

$$\pi = 19,012.50$$

- (b) (2 points). There is also a large group of local residents who would like to see the monument. Total daily demand from locals is:

$$Q^L(p) = 20,000 - 20,000p.$$

At the price you found in part (a), how many locals would visit the Taj Mahal?

Number of locals: 0

Answer:

No locals will visit.

$$Q^L(p) = 20,000 - 20,000 * 10.25 < 0$$

(c) (4 points). To promote access by the local community, the ADA moved towards charging local tourists – who have to show ID to qualify – a different admission price from the price it charges to foreigners.

Find the profit-maximizing price that the ADA should charge to local tourists.

Price: \$0.75

Answer:

$$Q^L(p) = 20,000 - 20,000p \rightarrow p = 1 - 0.00005Q^L$$

$$\pi = (1 - 0.00005Q^L)Q^L - 0.5Q^L$$

$$\frac{d\pi}{dQ^L} = 1 - 0.0001Q^L - 0.5 = 0 \rightarrow Q^L = 5,000, p(Q^L) = 0.75$$

How much additional profit does the ADA make from offering two different prices relative to the single-price strategy from part (a)?

Gain in Profit: \$1,250

Answer:

$$\pi = 5,000 * (0.75 - 0.5) = 1,250$$

Out of the total visitors to the Taj Mahal, what share is made up of local residents?

Share of locals: 71.94%

Price Discrimination (2)

The Pennsylvania Ballet is preparing for its performance of “The Nutcracker”, which attracts two different types of customers: one-time ballet visitors and ballet lovers. There are twice as many one-time visitors as there are ballet lovers. The marginal cost to the Ballet per ticket sold is \$20; assume that they have no capacity constraints and no preference for selling out any particular performance.

The Ballet conducted some market research and concluded that reservation prices for a single performance for the two groups are as follows:

	One-time Visitor (200 People)	Ballet Lover (100 People)
Afternoon performance (MC = \$20)	\$40	\$60
Evening performance (MC = \$20)	\$70	\$120

[2 points] The profit-maximizing ticket pricing strategy, if the Ballet chooses a single price, is to:

ANSWER: Offer a ticket to either performance for \$70:

Price (Either Show)	Profit
40	$(40-20)*300=$ 6,000
60	$(60-20)*300=$ 12,000
70	$(70-20)*300=$ 15,000
120	$(120-20)*100=$ 10,000

[3 points] The Ballet now considers charging different prices for afternoon and evening performances. The profit-maximizing choice has them:

Answer: price low good at \$40; high good at \$99.99.

Profits are $200*(40-20)+100*(99.99-20)=11,999$

15,000 > 11,999 Therefore, you would continue with the same single-price strategy you found above.

Risk Aversion

Target. Target offers insurance for their digital camera sales, so if the camera breaks within two years of purchase, the insurance holder can receive a new camera for free. A new Canon EOS Rebel T5i runs \$750 and has a 5% chance of breaking within the first two years of ownership.

- a) What is the actuarially fair price of this insurance plan?

Price = _____ \$37.50 _____

Solution:

The actuarially fair price is the expected cost, which is:

$$E(C) = 5\% \cdot \$750 = \$37.50$$

Target charges \$100 for the two-year insurance plan.

- b) How much expected profit does Target make on each insurance plan sold (assuming their replacement cost is actually \$750)? (Those of you who have shopped for electronics recently know that retailers must make a profit given how hard salespeople typically push these service plans.)

π = _____ \$62.50 _____

Solution:

$$E(\pi) = p - E(C) = \$100 - \$37.50 = \$62.50$$

Target would say this high price is due to risk aversion. Assume customers have a utility function given by $\ln(w)$ where w is wealth. NOTE: When dealing with logarithms, rounding too soon can lead to numerical errors. As such, don't round until the end.

- c) Would the consumer want the \$100 insurance plan if their wealth (after buying the camera the first time) were \$5,000 (assume they repurchase the camera if it breaks and they do not have insurance)?

YES

NO

Solution:

We just need to compare the expected utility of the customer when she buys the insurance relative to when she does not:

$$EU_{No\ Insurance} = 95\% \cdot \ln(\$5000) + 5\% \cdot \ln(\$5000 - \$750)$$

$$EU_{No\ Insurance} = 8.509$$

$$EU_{Insurance} = \ln(\$5000 - \$100) = 8.497$$

So she would not purchase the insurance.

- d) Assuming the **ln** utility function, what is this consumer's willingness to pay for the insurance plan? NOTE: To avoid rounding errors, when you compute any logarithms, please keep at least 6 digits after the decimal point.

$$WTP = \underline{\$40.47}$$

Solution:

We first find the certainty equivalent by comparing to the situation with no insurance:

$$EU_{No\ Insurance} = 95\% \cdot \ln(\$5000) + 5\% \cdot \ln(\$5000 - \$750) = 8.509067 = \ln(c)$$

$$c = e^{8.509067} = \$4959.53$$

$$c = w - \text{willingness to pay}$$

$$\$4959.53 = \$5000 - \text{willingness to pay}$$

$$\text{willingness to pay} = \$40.47$$

(Challenge Question) Critics say that the way Target justifies selling this insurance is by scaring customers into thinking that the probability of failure is much higher.

- e) What probability of failure in the first two years would justify a price of \$100 to someone who has wealth of \$5,000 (after buying the camera the first time)?

$$\text{Probability of Failure} = \underline{12.4\%}$$

We want the value of ρ from the following:

$$\ln(\$5000 - \$100) = (1 - \rho) \cdot \ln(\$5000) + \rho \cdot \ln(\$5000 - \$750)$$

Solving for ρ yields:

$$\rho = \frac{\ln(\$4900) - \ln(\$5000)}{\ln(\$4250) - \ln(\$5000)} = \frac{8.496990 - 8.517193}{8.354674 - 8.517193} = 0.124$$

So the probability of failure would need to be at least 12.4% to justify buying the insurance when their wealth is \$5,000.